

Goutam Mukku

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PROFESSIONAL EXPERIENCE

Tata Consultancy Services

Aug 2024 - Dec 2024

Machine Learning Engineer-Gen AI

Pittsburgh, PA

- Led team of 4 engineers, delivered Synthetic Data Generation using LLMs, recommending Claude-3.5 with 86% ML utility accuracy
- Conducted evaluations of GPT-4, Llama-2, Claude-3.5, Gemini, leveraging metrics like KL Divergence to identify best performing model
- Evaluated fidelity, privacy and utility of generated synthetic data using metrics membership inference score, Wasserstein Distance
- Validated prompt engineering techniques, with Chain-of-Thought improving statistical accuracy, reasoning, and diversity in SDG
- Documented entire ML Lifecycle, delivered KT session, handed over product documentation, ensuring seamless client adoption

Talentmapp

Jun 2024 - Aug 2024

Founding Product Manager-AI/ ML

Pittsburgh, PA

- Led user research and used Mixpanel to analyze behavior, prioritizing MVP features and enhancing acquisition by 20%
- Built the product roadmap for MVP, and prioritized 3 features for initial release, ensuring alignment with go-to-market strategy
- Researched and analyzed 4 third-party Data API providers, and finalized 1 provider that optimized data retrieval by 70%
- Supervised the vector embedding creation in MongoDB Atlas, and monitored accuracy testing achieving a 60% accuracy

Empowered Systems Pvt. Ltd

Nov 2021 - Apr 2023

Associate Solutions Consultant

Bangalore, IN

- Delivered end-to-end connected risk platform, generating \$1.1 Million in revenue and facilitating client training to drive adoption
- Led requirement-gathering session for 2 product features, producing comprehensive functional and non-functional requirements that accelerated implementation time by 45%
- Streamlined the internal audit approval workflow through feature development, slashing approval cycle from 336 hours to 12 hours
- Accelerated User Acceptance Testing completion in 2 weeks, improving resolution time by 50% and expediting product readiness
- Configured Connected Risk product for audit solutions and managed as client advisory in the risk management domain of 500+ employees in the USA, Canada, and Hong Kong region

Hewlett Packard Inc

Oct 2020 - Nov 2021

Data Scientist

Bangalore, IN

- Devised automated troubleshooting log generation, reducing repeat reseller service call costs and saving \$200,000 in annual support costs
- Built an ARIMA-based order forecasting model using SAP HANA SQL, Azure and PySpark, improving operational efficiency with 70% accuracy interactive dashboard
- Analyzed NLP-driven textual analysis using PySpark, SQL and Azure on JIRA tickets, reducing bug reports by 40%
- Spearheaded the user persona development, designed front-end technical specifications, and led UAT within an agile framework
- Supported lead scrum master with sprint planning, designed five product features, and managed JIRA to track user stories and bugs

EDUCATION

Carnegie Mellon University (CMU)

Aug 2023 - Dec 2024

Master of Information Systems Management

Pittsburgh, PA

Coursework: AI Ops, Application of LLM, Responsible AI, Statistics, Database Management, Agile Methods, Distributed Systems

Accomplishments: Dean's List Fall 2024, Teaching Assistant for Product Management Essentials, AI and Emerging Economies

Vellore Institute of Technology

Jul 2016 – Jul 2020

Bachelor of Technology, Computer Science

Vellore, IN

Coursework: Python, Database Management, Machine Learning, Natural Language Processing, Deep Learning

PROJECTS

- **AI-driven Traffic Prediction (AIOps):** Built and deployed a time-series traffic monitoring system using LSTM and GRU on Kubeflow with Evidently AI, where LSTM showed 58% fewer drifting features than GRU, enhancing model stability
- **Conversational AI Job search (Fine-Tuning):** Developed using Fine-Tuning techniques (LoRA and QLoRA), with LoRA achieving 63% improvement in METEOR for enhanced accuracy
- **Benefit assessment protocol for AI systems:** Developed a framework to evaluate stakeholders, measure trade-offs and quantify risks and benefits using capability-based approach, promoting Responsible AI practices
- **Customer Satisfaction Analysis:** Analyzed Customer sales data to assess 48-month Customer Lifetime Value (CLV), revealing high retention, with a key recommendation for new customer acquisition post-48 months to sustain revenue growth
- **AI-Powered Insights for Financial Consumer queries (NLP):** Applied NLP and OpenAI API to analyze sentiment in financial customer complaints and generate concise summaries, improving actionable insights

SKILLS

AI/ML: Regression, Classification, Clustering, PCA, Boosting, Bagging, RNN, ARIMA, SARIMA, LSTM, GRU, Attention Mechanism, Transformers

Generative AI: Prompt Engineering, RAG, Fine-Tuning, SFT, LoRA, QLoRA, Grafana, LLM Evals (BLeU, ROUGE, RAGAs, LLM-as-a-Judge)

MLOps: Azure Data Factory, AWS Sage Maker, Kubeflow, MLflow, Google Vertex AI, Docker, Kubernetes

Libraries: Scikit-learn, Tensorflow, PyTorch, Keras, XGBoost, NLTK, spaCy, NumPy, Pandas, HuggingFace, Langchain

Frameworks & Tools: Python, Big Data (Sql, MongoDB, Spark), Responsible AI (AIF360, SHAP), Power BI, Tableau, Git, A/B Testing

Management: Figma, Job-To-Be-Done, JIRA, Confluence, Requirement Analysis, Customer Satisfaction Score (CSAT), Monthly Active Users (MAU), Customer Churn Analysis, Competitor Analysis, Market Research

ACHIEVEMENTS

Presented an MVP Solution for Henkel at Princeton University Case competition, enhancing recycling engagement through gamification and cost-effective social integration; achieved a 2nd runner-up

CMU Heinz Consulting Club – Vice President– Organized a Case Competition with 100 participants and led a team of 12 for competition